Nahid Ahmadvand

Data Scientist

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SUMMARY

Data Scientist with an M.Sc. in Quantum Information and Computation and a background in Physics, focused on applied machine learning, NLP, time-aware validation, and model explainability. Built multiple end-to-end machine learning projects across sparse text modeling, customer churn prediction, web-session intrusion detection, Formula 1 analytics, anomaly detection, and sensor-data classification. Strong in leakage-aware experimentation, feature engineering, evaluation design, and translating model results into clear technical write-ups published in Medium data science publications.

TECHNICAL SKILLS

Languages Python, SQL

Libraries pandas, NumPy, SciPy, scikit-learn, LightGBM, SHAP, Gensim, Vowpal Wabbit, Matplotlib, Seaborn, Git, Jupyter, Hugging Face Datasets

Methods Classical NLP, TF-IDF, sparse matrices, Ridge/Logistic Regression/SVM/Random Forest/GBM, stacking, target encoding, time-aware cross-validation, conformal prediction, SHAP/PDP, error analysis

SELECTED PROJECTS

Medium Article Engagement Prediction - Leakage-Aware NLP + LightGBM *May 2026*

GitHub · Article

- Built a leakage-aware regression pipeline on 66,088 English Medium articles, modeling long-tailed clap counts with `log1p(claps)` and chronological train/validation/test splits to simulate future article prediction.
- Engineered a 238,671-feature sparse representation from body/title/metadata TF-IDF, author/publication categorical signals, and numeric article metadata; compressed sparse features with TruncatedSVD for LightGBM.
- Blended Ridge and LightGBM with validation-only calibration, reaching 1.0287 held-out chronological test MAE; ablation showed leakage-safe author/publication target encoding reduced validation MAE by 0.0718.

Intruder Detection Through Webpage Session Tracking - Kaggle *Mar 2026*

GitHub · Article

- Built a behavioral user-identification pipeline on 253,561 web sessions and 48,371 unique sites using sparse CSR matrices, CountVectorizer/TF-IDF site n-grams, timestamp features, and Logistic Regression.
- Applied chronological validation with TimeSeriesSplit and regularization tuning, achieving a best public Kaggle ROC-AUC of 0.94443.
- Diagnosed a preprocessing failure where unscaled session duration dropped leaderboard ROC-AUC to 0.66528, then corrected it with proper scaling to restore strong performance.

Spooky Author Identification - Classical NLP + Stacking *Jun 2026*

GitHub · Article

- Built a three-class authorship attribution pipeline on 19,579 literary sentences, engineering word n-gram, character n-gram, punctuation, and TF-IDF features to capture stylistic signals beyond topic keywords.
- Benchmarked Vowpal Wabbit, TF-IDF Logistic Regression, NB-SVM, Naive Bayes, SGD Logistic Regression, BM25, Word2Vec, FastText, and sparse-text ensembles.
- Stacked out-of-fold base-model predictions with a Logistic Regression meta-learner, improving holdout log loss from 0.3843 to 0.3504 and reaching 0.30414 private Kaggle log loss.

Customer Churn Prediction - Retention-Focused ML Workflow *Jan 2026*

GitHub · Article

- Built a leakage-safe churn prediction pipeline on 7,043 Telco customers using ColumnTransformer, stratified cross-validation, class weighting, and model benchmarking across Logistic Regression, KNN, Decision Tree, and Random Forest.
- Tuned Random Forest achieved 0.846 ROC-AUC and 0.789 churn recall; top-10% targeting reached 75.9% precision and 2.86x lift over random outreach.
- Applied SHAP and permutation importance to identify churn drivers such as contract type, tenure, support/security services, payment method, and billing features.

Ferrari 2025 Expected vs Actual Points - F1 Performance Audit *Mar 2026*

GitHub · Article

- Built a reproducible Formula 1 expected-points audit using 2018-2024 race, sprint, qualifying, standings, circuit, and weather-climatology data to score Ferrari's 2025 season out of sample.

- Engineered leakage-safe pre-session features and selected an ElasticNet model with time-aware cross-validation, achieving 2.695 MAE and 4.226 RMSE per session.
- Added split conformal prediction intervals and a diagnostic attribution ledger, quantifying a 91.92-point net deficit and decomposing 158.53 clipped lost points into reliability, qualifying-conversion, and residual buckets.

F1 SC/VSC Onset Forecastability Triage - Time-Series Diagnostics *2026*
GitHub

- Built a lap-level Safety Car / Virtual Safety Car onset forecasting framework with event-based temporal splits, leakage controls, calibration checks, and event-level evaluation.
- Engineered seven race-context feature families and added forecastability triage, reconstructing a cleaner next-lap target to show that useful signal was mainly covariate-driven rather than target-memory-driven.

ADDITIONAL PROJECTS

- **Human Activity Recognition:** Compared K-Means/Ward clustering vs. LinearSVC on 10,299 smartphone sensor samples; PCA retained 90% variance in 65 components, while full-feature LinearSVC achieved 0.96 held-out macro F1.
- **Sarcasm Detection:** Built a leakage-safe TF-IDF + Logistic Regression pipeline on 1M+ Reddit comments, removing explicit sarcasm markers and achieving approx. 0.71 F1, 0.79 ROC-AUC, and 0.80 PR-AUC.
- **Wine Quality Prediction:** Compared OLS, Lasso, and Random Forest on 4,898 white wines; tuned Random Forest reduced holdout RMSE from 0.7644 to 0.6048 and used permutation importance, PDP, and SHAP for interpretation.
- **Olivetti Faces:** Built PCA/eigenfaces, K-Means, GMM, Random Forest, and reconstruction-error anomaly detection pipeline; reduced 4096 pixels to 199 PCA components and achieved 97.5% test accuracy with PCA features.
- **Decision Tree from Scratch:** Implemented a scikit-learn-compatible decision tree supporting Gini, entropy, variance, MAD-median, `predict`, and `predict_proba`; reached 88.33% test accuracy on handwritten digits.

EDUCATION

M.Sc. Quantum Information and Computation *Sep 2021 – Feb 2024*
Sharif University of Technology, Tehran, Iran — GPA: 18.55 / 20

- Honorary direct admission to Sharif's M.Sc. program without the nationwide graduate entrance exam, awarded after graduating ranked 1st by GPA in B.Sc. Physics.
- Master's thesis research published in *Physical Review A*: Exact quantum noise deconvolution with partial knowledge of noise.

B.Sc. Physics *Sep 2017 – Jul 2021*
University of Tehran, Tehran, Iran — GPA: 18.15 / 20 — Ranked 1st by GPA

WRITING

- Publish reproducible machine learning project write-ups on Medium across NLP, sports analytics, tabular modeling, unsupervised learning, anomaly detection, and explainability, with articles published in *Data Science Collective*, *Formula One Forever*, *Data & Beyond*, and *Python in Plain English*: medium.com/@nahid.nm57.

CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI / Stanford Online via Coursera, Aug 2025.